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Business Mathematics

Week 8: Module 11

GOVERNMENT PROPERTY
NOT FOR SALE

Business Mathematics

Week 8: Module 11

Target

Some businesses encounter neither a profit nor a loss. Imagine when you sell products and your total revenue of the business is equal to its total costs. In this case, you did not make a profit and no loss occurred. It is somewhat sacrificing your time and effort selling goods ending up with nothing. This point of not making profit and no occurring loss is called break-even.

Why do businesses encounter break-even? Is it okay for a business to break-even? What shall you do when you knew the break-even point? How can you determine break-even point? These are the questions in our minds about break-even.

This learning material will immerse you with information and activities that will help you understand break-even point. After going through this learning material, you are expected to determine the break-even point.

Before going on, check how much you are aware of the words used in the lesson. Do the game on the next page.

Module 13	Break - even Point

In the previous modules, you have learned about pricing, profit and loss, and the discount. These topics talk about the processes and examples in buying and selling. In this module, you will learn about break- even point and how this topic will help you understand buying and selling.

Before going on to this module, kindly answer the preliminary test below.

Pre-Test:

Your score in this activity ***WILL NOT BE RECORDED*** *but you are expected to answer the questions as honest as you can. This will serve as the preliminary assessment that could be helpful in evaluating you by your subject teacher.*

DIRECTIONS: Read carefully each item. Choose the letter of the best answer for each test item. Use a separate sheet for your answers.

1. Which of the following are not used in break-even point analysis?
   I. Account statement
   II. Billing statement
   III. Income statement
   IV. Statement of Assets, Liabilities and Net Worth
   A. II, III, IV B. I, III, IV C. I, II, IV D. I, II, III
2. Which of the following is obtained from multiplying selling price per unit by number of units?
   A. Break-even point C. Total cost
   B. Contribution margin D. Total revenue
3. It is a type of expenses that do not increase when number of units produced increases.
   A. Contribution margin C. Fixed Cost
   B. Discount D. Variable cost
4. Which of the following is NOT used in break-even point analysis?
   A. Contribution margin C. Fixed Cost
   B. Discount D. Variable cost
5. Which of the following mathematical statement shows MORE profitable?
   TR means total revenue and TC means total cost.
   A. TR = TC B. TR < TC C. TR > TC D. TR ≥ TC
6. Which of the following are NOT classified as fixed cost?
   I. Advertising of product
   II. Piece rate labor
   III. Raw materials
   IV. Rent
   A. II, III, IV B. I, III, IV C. I, II, IV D. I, II, III

Discover

In the previous activity, you have learned that there are words like revenue and cost linked to break-even. Let us know more about these words with the help of the income statement below.

The income statement below shows the results of operation of Ben Dilan Photoshop. This business develops pictures of the same size and sell them with a reasonable price.

BEN DILAN PHOTOSHOP
Income Statement
For the Year Ended December 31, 201A

Sales (15,000 pictures at ₱19.00)		₱285,000
Less		
Variable costs of goods sold (15,000 x ₱8.00) ...	₱120,000	
Variable selling costs (15,000 x ₱2.00)	₱30,000	₱150,000
Contribution Margin		₱135,000
Less		
Fixed Costs		₱8,001
Operating Profit		₱126,999

Total revenue refers to the total sales based on the total number of units of outputs produced and sold. This may also refer to gross sales because it is located at the first line of the income statement proper, labeled as Sales.

By formula,

$$TR = P \cdot X$$

where TR is the Total Revenue,
P is the Average selling price per unit (P), and
X is the number of Units of output produced and sold.

Example: Ben Dilan Photoshop

Total revenue = Selling price per unit x number of units sold
= ₱19.00 x 15,000
= ₱285,000

Expenses, on the other hand, are costs spent to make the revenue. Expenses may refer to *variable cost* and *fixed cost.*

Variable cost (V) is the amount of expenses that is directly proportional to the number of units produced and sold (x). As the number of units increases, so does the variable cost. Examples of variable costs are raw materials, sales commissions, delivery fee or shipping fee, advertisements, and the like. Compute variable costs in Ben Dilan Photoshop.

Variable costs of goods sold = Variable cost per unit x number of units
= ₱8.00 x 15,000
= ₱120,000
Variable selling costs = Variable cost per unit x number of units
= ₱2.00 x 15,000
= ₱30,000

Variable costs of goods sold may refer to amount spent for raw materials to create pictures and variable selling costs may refer to delivery fee.

Total Variable cost = Variable costs of goods sold + Variable selling costs
= ₱120,000 + ₱30,000
= ₱150,000

Fixed cost (FC), on the other hand, is the amount of expenses that do not change regardless of the increase or decrease in the number of units produced and sold. Examples of fixed costs are rent for office space, weekly payroll, amortization, and the like. Unlike Variable cost, there's no further computation for fixed cost. For instance, the monthly rent for office space in Ben Dilan Photoshop is ₱8,001, this is already the fixed amount that the owner pays every month.

So, total cost is the sum of total variable cost and the total fixed cost

Total cost = Total Variable Cost + Fixed Cost
= ₱150,000 + ₱8,001
= ₱158,001

Alternatively, the **Total Cost Formula** can be expressed as follows

$$\boxed{TC = (V \cdot X) + FC}$$

where TC is the Total Cost,
V is the Variable cost per unit,
X is the number of Units of output, and
FC is the Fixed Costs.

ALTERNATIVE SOLUTION:

$$TC = (V \cdot X) + FC$$
$$TC = (₱10 \times 15{,}000) + ₱8{,}001$$
$$TC = ₱150{,}000 + ₱8{,}001$$
$$TC = ₱158{,}001$$

If the total revenue is greater than the total costs, then there is a profit. And if the total revenue is less than the total costs, you lose. In Ben Dilan Photoshop, its total sales is greater than the total cost so it has operating profit of ₱126,999.00 *(this profit is obtained by subtracting the total cost from the total revenue).*

Profit may be calculated by subtracting the total costs from the total revenue. In symbols,

$$Profit = TR - TC$$
$$Profit = (P \cdot X) - [(V \cdot X) + FC]$$
$$Profit = (P \cdot X) - (V \cdot X) - FC$$

$$\boxed{Profit = [X(P - V)] - FC}$$

However, if the Total Revenue equals Total Cost, then the business is said to be in **Break-even**. This means that the business neither make a profit nor loss. To determine the break-even point, calculate the number of units of output that makes total revenue equal to total cost. Now, since there is no profit and no loss, we can use zero (0) as a substitute.

$$Profit = [X(P - V)] - FC$$
$$0 = [X(P - V)] - FC$$

Simplifying this further,

$$X(P - V) = FC$$

BREAK- EVEN POINT FORMULA

$$X = \frac{FC}{(P - V)}$$

NOTE: *If the math problem uses variables X, FC, P, and V, you can use the following formulas if one variable is known and the other variables are unknown.*

Finding the value of FC when X, P, and V are known	**Finding the value of P** when X, FC, and V are known	**Finding the value of V** when X, FC, and P are known
$FC = X(P - V)$	$P = \frac{FC}{X} + V$	$V = P - \frac{FC}{X}$

where V is the Variable cost per unit,
P is the selling price per unit,
X is the number of Units of output, and
FC is the Fixed Costs.

The $(P - V)$ in the break-even formula is called **unit contribution margin**. This is the difference between average selling price per unit and variable cost per

unit. It can be referred to as the leftover to cover fixed costs and then add to operating profit.

Consider the sample income statement of Ben Dilan Photoshop in computing the contribution margin:

SOLUTION A:

Unit Contribution Margin = Selling Price (P) – Total Variable Cost per unit (V)
= ₱19.00 – (₱8.00+₱2.00)
= ₱19.00 - ₱10.00
= ₱9.00

SOLUTION B:

Unit Contribution Margin = [Total Revenue (TR) – Total Variable Cost (V)] ÷ Total Number of Units Produced and Sold
= (₱285,000 - ₱150,000) ÷ 15,000
= ₱135,000 ÷ 15,000
= ₱9.00

To compute the break-even Point in Units:

BEP in Units (X) = Fixed Cost (FC*)* ÷ Unit Contribution Margin (P – *V)*

$$X = \frac{FC}{(P - V)}$$

$$= \frac{₱8,001}{₱9.00}$$

= 889 pictures

Interpretation: This means that if you create and sell 889 pictures, you cannot make a profit and will not incur any loss. Moreover, production of less than 889 pictures will incur loss in the business while production of more than 889 pictures will make profit. Also, an increase of ₱9.00 (*the unit contribution margin*) in the operating profit will be added for every picture in excess of 889. For example, the owner created and sold 889 pictures, the operating profit will be ₱0, 890 pictures will give ₱9.00, 891 pictures will give ₱18.00, and so on and so forth.

To compute the break-even Point in Pesos:

BEP in Pesos = BEP in units (X) x Selling price per unit (P)
= 889 × ₱19.00
= ₱16,891.00

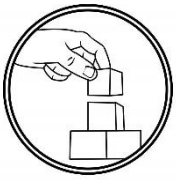
Interpretation: This means that if you created 889 pictures and sold it for ₱16,891.00, you did not make any profit and no incurred loss. So, what will you do now in order to make profit? You have to increase the production or increase the number of pictures to be sold.

Let us prove that break-even in this example can make a zero profit using 889 pictures only.

BEN DILAN PHOTOSHOP
Income Statement
For the Year Ended December 31, 201A

Sales (889 pictures at ₱19.00)		₱16,891
Less		
Variable costs of goods sold (889 x ₱8.00)	₱7,112	
Variable selling costs (889 x ₱2.00)	₱1,778	₱8,890
Contribution Margin		₱8,001
Less		
Fixed Costs		₱8,001
Operating Profit		0

Break-even Point Analysis is a guide for every business owner to plan for the target profit or the target sales. Nobody wanted to operate business without profit. Business is all about creating strategies that could lead to a more profitable outcome. Thus, the examples presented in this lesson helped you in the decision-making process as to how should a business prioritize production.



Explore

Enrichment

Here are some enrichment activities for you to work on to master and strengthen the basic concepts you have learned from this lesson.

Activity 1: Is it a Fact or a Bluff?

Directions: Read carefully each item and tell whether it is a fact or a bluff. Write FACT if the statement is true, otherwise write BLUFF. Use a separate sheet for your answers.

1. The variable X in the break-even formula refers to the number of items produced and sold.
2. The operating profit in the break-even point is ₱1.00.
3. Total revenue is also known as the total sales.
4. Fixed cost is an example of expenses.
5. There are two types of expenses namely, variable cost and fixed cost.
6. Variable cost is the amount that doesn't change despite of the increase in the number of items.
7. Total cost is the sum of the total variable cost and total fixed cost.

8. Profit is obtained by subtracting the total revenue from the total cost.
9. Unit contribution margin is obtained from multiplying the selling price per unit by the number of units or items produced.
10. Break-even point in units refers to the number of items produced such that the operating profit is 0.

Enrichment Activity 2: Complete me!

Directions: Fill in the needed data to complete the table below.

	Total Fixed Cost	Selling Price per Unit	Variable Cost per Unit	Unit Contribution Margin	B.E.P. in units	B.E.P. in Pesos
1	₱370.00	₱14.00	₱2.00			
2	₱1,600.00	₱20.00	₱4.00			
3	₱19,500.00	₱15.00		₱13.00		
4	₱960.00	₱18.00		₱15.00		
5		₱24.00	₱3.00		69	
6		₱34.00	₱3.00		150	
7		₱27.00	₱5.00		100	
8		₱9.00		₱7.00		₱2,700.00
9		₱11.00		₱8.00		₱1,562.00
10		₱15.00		₱11.00		₱1,185.00

Enrichment Activity 3: Accept the Math Challenge!

Directions: Read carefully the word problem below to complete the income statement and answer the questions that follows.

Mr. Juan Dilo-dilo is the owner of a convenient store. One of his products is sugar where he bought 25 kg worth ₱60.00/kg. His 2 employees will repack the sugar by ¼ kg in a plastic bag, so this will give 100 plastic bags of sugar. Each sugar in plastic bag will be worth ₱25.00 each. The 2 employees are given a day to finish repacking. Other expenses for sugar repacking are as follows:

Cost of a plastic bag = ₱0.90
Labor cost in a daily basis = ₱250.25 per employee

Academic Senior High School of the City of Meycauayan
St. Francis (Gasak), City of Meycauayan, Bulacan
Subject Teacher: **Mrs. Lailah Claire U. Ronidel**
Contact Number: **09684694646**
Facebook Messenger Account: **Ma'am Lailah Ronidel**

BUSINESS MATH QUARTER 3

Most Essential Learning Competencies	Output To Be Submitted	Date of Completion
Quarter III Week 1 ***2 Fractions/ decimals and percentage*** **February 21-25, 2022** ✓ Give real-life situations to illustrate fractions, decimals, and percent (ABM_BM11FO-Ic-4) ✓ Solve problems involving fractions, decimals and percent (ABM_BM11FO-Id-5)	**Worksheet for Week 1 -2 (Reference: Activity Sheet Quarter III)** Instructions: Answer the following activities. Module 1-2 . Gauge. Activity 8. Convert Me! (A, B, C, D, E, F) o Write all your answers on yellow paper. o Write the activity title o Write the quarter number and week number on the top right side of the paper.	**March 3, 2022**
Quarter III Week 3 – ***Key Concepts of Ratio and Proportion*** **February 28 – March 4, 2022** ✓ Identify the different kinds of proportions and write examples of real-life situations for each	**Worksheet for Week 3 (Reference: Activity Sheet Quarter III)** Instructions: Answer the following activities. Module 3. Explore. Activity 1. #s 1-5	**March 7, 2022**
Quarter III Week 4 ***– Key Concepts of Ratio and Proportion*** **March 7-11, 2022** ✓ Solve problems involving direct, inverse and partitive proportion (ABM_BM11RP-If-4)	**Worksheet for Week 4 (Reference: Activity Sheet Quarter III)** Instructions: Answer the following activities. Module 4. Gauge #s 1-15	**March 14, 2022**
Quarter III Week 5 ***Key Concepts of Buying and Selling*** **March 14-18, 2022** ✓ Differentiate Mark-on, Mark-down, and Mark-up obtain Mark-on, Mark-down and Mark-up of a product (ABM_BM11BS-Ig-1)	**Worksheet for Week 5 (Reference: Activity Sheet Quarter III)** Instructions: Answer the following activities. Module 5a. Explore. Activity 1. Can You Tell? #s 1-10 Module 5b, Explore. Activity 1. Complete Me! Letters – A, B, C	**March 21, 2022**
Quarter III Week 6 ***Key Concepts of Buying and Selling*** **March 21-25, 2022** ✓ Differentiate Mark-up from margins (ABM_BM11BS-Ih-3) ✓ Describe how gross margins is used in sales (ABM_BM11BS-Ih-4) ✓ Compute single trade discounts and discount series	**Worksheet for Week 6 (Reference: Activity Sheet Quarter III)** Instructions: Answer the following activities Module 6. Gauge #s 1-10 Module 7. Gauge #s 1-10 Module 8. Enrichment. Activity 1. Solve Me #s 1-3	**March 28, 2022**

Quarter III Week *7 Key Concepts of Buying and Selling* **March 28- April 1, 2022** ✓ Differentiate profit from loss (ABM_BM11BS-Ii-6) ✓ Illustrate how profit is obtained and how to avoid loss in a given transaction (ABM_BM11BS-Ii-7)	**Worksheet for Week 7 (Reference Activity Sheet Quarter III)** Instructions: Answer the following activities Module 9. Explore #s 1-5 Module 10. Gauge #S 1-10	**April 4, 2022**
Quarter III Week *8 Key Concepts of Buying and Selling* **April 4-8, 2022** ✓ Determine the break-even point ✓ Solve problems involving buying and selling products (ABM_BM11BS-Ij-9)	**Worksheet for Week 8 (Reference: Activity Sheet Quarter 1II)** Instructions: Answer the following activities Module 11. Explore. Activity 1. Is it a Fact or a Bluff?	**April 11, 2022**
Quarter III Week 9 Key Concepts of Buying and Selling **April 11-13, 2022** ✓ Compute interest specifically as applied to mortgage, amortization and on services/ utilities and on deposits and loans	**Worksheet for Week 8 (Reference: Activity Sheet Quarter 1II)** Instructions: Answer the following activities Module 13. Gauge # 1-15	**April 18, 2022**